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ANLY 500 - Week 02 Lab Grading Rubric

Assignment: Introduction to Data Analytics 1 - HW2

Total Points: 100

Passing Grade: 70 (C)

Part 1: Variables, Hypothesis, Designs (40 points)

Students answer conceptual questions based on the research abstract about offshore outsourcing.

Question	Points	Grading Criteria
1. Hypothesis	5	Student identifies a clear, testable hypothesis related to outsourcing effects on cost savings, job creation, or economic impact.
2. Independent Variable	3	Student correctly identifies an IV (e.g., outsourcing location, where jobs are sent, country choice).
2a. IV Type	2	Student correctly classifies IV as nominal or categorical . Partial credit (1 pt) if mentioned but not explicitly stated.
3. Dependent Variable	3	Student correctly identifies a DV (e.g., cost savings, economic impact, job numbers, business success).
3a. DV Type	2	Student correctly classifies DV as ratio , continuous , or interval . Partial credit (1 pt) if mentioned but not explicitly stated.
4. Measurement Error	5	Student discusses potential sources of measurement error (e.g., currency conversion, reporting accuracy, calculation methods, self-report bias).
5. Research Design Type	3	Student identifies study as observational , correlational , survey-based , or descriptive .
5a. Design Justification	2	Student explains rationale (e.g., "no manipulation of variables," "uses existing data," "no random assignment"). Partial credit (1 pt) for brief/incomplete justification.
6. Reliability Measurement	5	Student explains how to assess DV reliability (e.g., test-retest reliability, internal consistency, Cronbach's alpha, repeated measurements).
7. Ecological Validity	5	Student assesses whether findings generalize to real-world settings and provides reasoning.
8. Cause & Effect	3	Student states whether causation can be claimed (answer should be "no" or "limited" for observational design).
8a. Causation Rationale	2	Student explains why causation cannot be established (e.g., "no manipulation," "confounding variables,"

Question	Points	Grading Criteria
		"observational data"). Partial credit (1 pt) for incomplete explanation.
9. Data Collection Type	3	Student identifies collection method as secondary data , archival data , or existing data .
TOTAL PART 1	40	

Part 1 Grading Notes

- All questions require **written responses** demonstrating understanding of research methodology concepts
- No R coding required in Part 1
- Award partial credit for responses that show understanding but lack complete detail
- Deduct points for missing key terminology or incomplete reasoning

Part 2: R Programming & Data Analysis (60 points)

Students perform statistical analysis using R on the `03_lab.csv` dataset containing outsourcing data.

Question	Points	Type	Grading Criteria	Interpretation Required?
1. Frequency Table	10	Code	Student uses <code>table()</code> function to create frequency table of Jobs variable. Code must execute and produce valid output.	No
2. Histograms	10	Code	Student creates histograms for both Cost and Cost2 using	No

Question	Points	Type	Grading Criteria	Interpretation Required?
			<code>hist()</code> function with <code>breaks=15</code> parameter. Deduct 3 pts if breaks parameter missing.	
3a. Most Normal Distribution	5	Interpretation	Student provides written interpretation identifying which cost savings variable appears most normal and explains why (e.g., "symmetric," "bell-shaped," "centered around mean").	YES - 50% penalty if missing
3b. Multimodal Distribution	5	Interpretation	Student provides written interpretation identifying which variable is multimodal and explains why (e.g., "shows multiple peaks," "two distinct modes").	YES - 50% penalty if missing
3c. Skewed Distribution	5	Interpretation	Student provides written interpretation identifying which variable is skewed, specifies direction (positive/negative or right/left), and explains (e.g., "long tail extending right").	YES - 50% penalty if missing
3d. Kurtotic Distribution	5	Interpretation	Student provides written interpretation identifying which variable shows kurtosis and explains (e.g.,	YES - 50% penalty if missing

Question	Points	Type	Grading Criteria	Interpretation Required?
			"heavy tails," "peaked center," "leptokurtic").	
4. Z-score Calculation	10	Code	Student uses <code>scale()</code> function or manual calculation <code>(x - mean(x)) / sd(x)</code> to compute z-scores for both cost savings variables.	No
6. Extreme Values	10	Code + Answer	Student identifies and counts values with z-scores beyond ± 1.96 ($p < .05$). Must provide numeric counts for both variables. Deduct 5 pts if counts not provided.	No
7. Min/Max Business IDs	10	Code + Answer	Student identifies business IDs with highest and lowest cost savings using both cost columns. Must provide specific ID numbers .	No
TOTAL PART 2	60			

Part 2 Critical Grading Rule: 50% Penalty for Missing Interpretations

Questions 3a-3d explicitly require written interpretations of histogram visualizations.

Full Credit (5 points per question)

Student provides:

- ☒ Correct identification of the distribution
- ☒ Written explanation using appropriate statistical terminology
- ☒ Reference to visual features observed in the histogram

Example: *"Cost Savings 1 appears most normal because it displays a symmetric, bell-shaped distribution with data centered around the mean and no obvious skewness."*

50% Penalty (2-3 points per question)

Student provides:

- ☒ Histogram code that executes correctly
- ☒ Visual output showing the distribution
- ☒ **NO written interpretation or explanation**

Example: *[Shows only code: `hist(data$Cost, breaks=15)` with graph output but no text explanation]*

Zero Credit (0 points)

- No code provided
- Code does not execute
- Interpretation is completely incorrect
- Question left blank

Why the 50% Penalty?

Lab instructions state: **"Examine these histograms to answer the following questions:"**

This requires students to:

1. **Generate** visualizations (coding skill)
2. **Interpret** distributions (statistical literacy)
3. **Communicate** findings (scientific writing)

Simply showing a graph demonstrates only 1 of 3 competencies.

Grade Scale

Letter Grade	Point Range	Percentage
A	90-100	90-100%
B	80-89	80-89%
C	70-79	70-79%
D	60-69	60-69%
F	0-59	Below 60%

Common Deductions

Part 1

- **-1 to -2 pts:** Variable type not clearly stated (IV/DV classification)
- **-1 pt:** Design justification too brief ("because it's observational" without elaboration)
- **-3 pts:** Missing data collection type entirely
- **-5 pts:** No discussion of measurement error sources
- **-5 pts:** Missing reliability or ecological validity assessment

Part 2

- **-3 pts:** Histogram missing **breaks=15** parameter
 - **-2.5 to -3 pts per question:** Missing interpretation on Q3a-d (50% penalty)
 - **-5 pts:** Code provided but extreme value counts not stated
 - **-5 to -10 pts:** Code does not execute or produces errors
 - **-10 pts:** Entire question missing (Q1, Q2, Q4, Q6, Q7)
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Grading Workflow

Step 1: Verify Submission Format

-  Submitted as Word document (.docx) or HTML
-  Student name in author field
-  Code chunks executed and showing output
-  All questions addressed

Step 2: Grade Part 1 (40 points)

- Read each written response
- Check for key concepts and terminology
- Award full, partial, or zero credit per question
- Note specific feedback for improvement

Step 3: Grade Part 2 (60 points)

- Verify code executes and produces correct output
- For Q1, Q2, Q4, Q6, Q7: Award full credit if code/output present
- **For Q3a-d:** Check for written interpretations
 - Full credit: Interpretation present
 - 50% penalty: Code only, no interpretation
 - Zero credit: Nothing provided
- Check numeric answers are provided for Q6-Q7

Step 4: Calculate Total & Letter Grade

- Sum Part 1 + Part 2 scores
- **Maximum:** Part 1 = 40, Part 2 = 60, Total = 100
- Assign letter grade per scale above

Step 5: Provide Feedback

- List specific areas of strength
- Identify questions with missing interpretations
- Suggest improvements for future assignments

Academic Integrity Note

Students may use:

-  Course lecture materials and textbook
-  R documentation and help files
-  AI assistants for concept clarification (ChatGPT, Claude, Copilot)
-  Discussion with classmates about general concepts

Students may NOT:

-  Copy code directly from other students
-  Submit work completed by another person
-  Use solutions from previous semesters
-  Plagiarize written explanations

All sources must be properly cited in APA format.

Rubric Version

- **Version:** 1.0
- **Last Updated:** December 18, 2025
- **Course:** ANLY 500 Analytics I
- **Instructor:** Ziyuan Huang